10 The Transition to Formal Knowledge

This chapter considers the challenges that students face as they make the transition from the embodiment and symbolism of school mathematics to formal mathematics at university based on set-theoretic definition and formal proof. The framework of three worlds of mathematics will be used to analyze supportive and problematic aspects of fundamental definitions including sets, relations, functions, equivalence relations and order relations. There follows a review of the formal construction of the real numbers and how students attempt to make sense of the concept of limit in mathematical analysis. This reveals a spectrum of different personal conceptions from 'natural' approaches based on previous meanings of embodiment and symbolism to a 'formal approach' based on making deductions from formal set-theoretic definitions. The chapter analyses how some students 'give meaning' to the definition from their embodied and symbolic experience, and some 'extract meaning' directly from the definition by learning to formulate formal proofs from formal definitions, with varying levels of confidence and disaffection, depending on the supportive and problematic aspects of their development. Various different methods of analysis of data are considered and related to the long-term framework for mathematical thinking presented in this book.

1. Major Changes from Embodiment and Symbolism to Formalism

The shift from the practical and theoretical development of embodied objects and symbolic operations to the formalism of axiomatic systems involves significant changes in meaning. A student who has studied Euclidean geometry has experience of building a knowledge structure where theorems are deduced from axioms and previous theorems. However,

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this involves principles such as congruence or the properties of parallel lines rather than the quantified set-theoretic deductions in formal proof. Meanwhile, proof in arithmetic and algebra, based on the 'rules' of arithmetic, is also a useful prelude to formal proof from axioms.

School mathematics is based on embodied operations in geometry or arithmetic that arise naturally in practical situations, evolving into theoretical definitions and deductions that evolve from practical experiences. The shift to an axiomatic formal approach is likely to be problematic because formal operations do not require any procedure of operation to be specified. All that is required is that for two elements x, y in a given set, there is a third element written as $x \circ y$ that satisfies properties such as $x \circ y = y \circ x$.

In a formal approach, it is essential to focus on the properties of an operation, without needing to say how to define it. In this way, all formal proofs deduced from specific axioms and set-theoretic definitions will work in *any* system satisfying those axioms and definitions, regardless of the particular procedure used to perform the operation.

The terms 'natural' and 'formal' are consistent with their use in history, where 'natural philosophy' and 'natural science' were used to describe the study of natural phenomena using geometry, arithmetic, algebra and calculus before the introduction of Hilbertian formalism.

Given the evident power of formal proof, it is the mode of operation for presenting research mathematics by experienced mathematicians. But the changes in meaning required in passing from earlier embodied and symbolic experience to formal reasoning cause the transition to axiomatic formalism to be problematic for many undergraduates. The origins of these difficulties lie in our met-befores, which may involve implicit properties that are not explicitly specified in the formal definition.

2. Sets and Relations

The challenges underlying a formal approach to mathematics may be illustrated by a study of the foundational ideas of sets and relations.

The transition to formal mathematics is built on the concept of set. This is not given a formal definition (at least not until later developments in logic are introduced). Therefore students must build on their familiar experience to make sense of it. Fundamentally, most mathematicians recognize a set S as something that has elements, or members, so that for any x whatever, it can be determined whether x is a member of S (written $x \in S$) or not ($x \notin S$).

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Building on the implicit notion of set, various new constructions can be introduced, such as the set $A \times B$, which is defined to be the set of all ordered pairs (x, y) where $x \in A$ and $y \in B$. Each element in $A \times B$ relates to precisely one element in A and one element in B. That is all. No more assumptions need to be made. However, in the case of the set $\mathbb{R}$, which can be embodied by the number line, the set $\mathbb{R} \times \mathbb{R}$ can be visualized in terms of points in the plane $\mathbb{R}^2$. The mental picture may be less easily visualized for more general sets such as $\mathbb{R}^2 \times \mathbb{R}^3$ or $\mathbb{C} \times \mathbb{C}$.

A *relation* between two sets A, B is defined to be *any* specified subset R of $A \times B$. That's it. There is no restriction placed on what kind of subset it might be. However, specific types of relations – such as functions, order relations and equivalence relations – may be conceptualized in individual ways that are affected by the student's previous experience. As a consequence, students may imagine relations as having different implicit properties from those intended by the definitions, as we shall now see.

2.1 *Functions*

A *function* from a set A to a set B is a relation F that satisfies two further properties:

(F1) For each $x \in A$ there exists an element $y \in B$ such that $(x, y) \in F$.
(F2) If $(x, y_1), (x, y_2) \in F$ then $y_1 = y_2$.

Properties (F1) and (F2) together say that for each $x \in A$ there is precisely *one* element in $x \in B$ such that $(x,y) \in F$. The element y may be written as $F(x)$, enabling us to imagine the set F as the *graph* of the function consisting of all ordered pairs $(x, F(x))$ for $x \in A$. The graph is written as $F : A \to B$, where A is called the domain and B the target (or co-domain).

In the case where this is embodied as the graph of a real function $F : A \to \mathbb{R}$ (where the domain A is a subset of $\mathbb{R}$) then (F1) tells us that there is a point $F(x)$ such that $(x, F(x))$ is on the graph and (F2) tells us that this is the only one. It gives 'the vertical line test' to check if a relation is a function. (Figure 10.1, overleaf.) Draw a vertical line through any point $x \in A$ to see that it meets the graph in just one point, which will be $(x, F(x))$.

Apart from this, the formal definition makes no further restriction on functions.

The functions met in school mathematics have various familiar features that affect the meaning of the function concept. For instance, a function is usually given by a formula made up of polynomials, trigonometric

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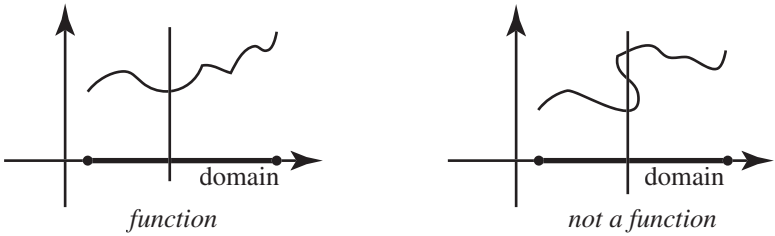
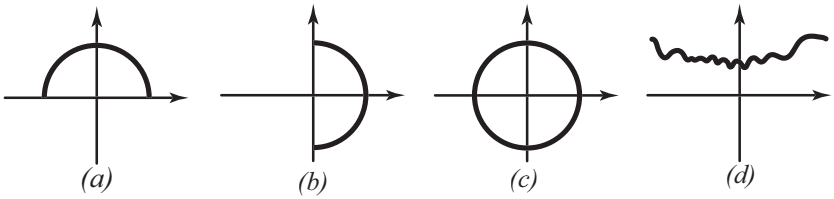


Figure 10.1. The vertical line test.

functions, exponentials and logarithms – all of which have recognizable shapes. In the transition from school to university, these met-befores may suggest implicit properties that a function must have, for instance, it is usually given by a single formula and students rarely have experience of a function having different formulae on different parts of the domain.

In a study of 36 students in the last year of secondary school and 109 first-year university mathematics students, MdNor Bakar asked which of a number of pictures represented functions.[1] (Figure 10.2.)



	%yes	%no
school	**61**	36
university	**91**	9

	%yes	%no
school	43	**57**
university	70	**28**

	%yes	%no
school	64	**29**
university	65	**35**

	%yes	%no
school	**14**	72
university	**72**	26

Figure 10.2. Which of the sketches could represent functions? Give an explanation.

Graph (a) was problematic for school students, with several saying 'if it were a function, the graph would continue, not stop,' or 'functions are usually continuous, needs a condition.' The term 'continuous' here refers to the dynamic continuity of the embodied action of drawing the curve with the expectation that it will continue to flow and not stop suddenly, which is characteristic of most examples in school mathematics.

[1] Bakar & Tall (1992).

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There is also an implicit assumption in the question that the pictures represent y as a function in x, so it was expected that students would say that picture (b) is not a function. However, two university students who declared it was a function commented in ways that suggested that x is a function of y, one writing 'look at it a different way', the other writing the equation as $f(y) = x$.

Two thirds considered (c) to be a function, which is perhaps not surprising given that it is sometimes described as an 'implicit function', even though it doesn't satisfy the vertical line test.

Graph (d) was rejected by three quarters of the students in school and a quarter at university. Their reasons included 'no regular pattern,' 'too irregular,' 'too complicated to be a function.' Experience in school refers to familiar functions given by an explicit formula, and the graph did not fit their mental image.

These problematic aspects may be addressed directly specifically by encouraging students to reflect on the precise meaning of the formal definition. For instance, it is essential to focus on what the definition says and what it doesn't say. Axiom (F1) says for each x there is a y, but it doesn't say that different xs must have different ys. Nor does it say that every y must be related to some given x. Nor does it say that the function must be given by a specific formula. By considering a range of examples illustrating different aspects of the function concept, a professor as mentor may assist students to refine their personal conceptions.[2]

2.2 *Relations between Elements in the Same Set*

When the two sets A and B are the same, we speak of *a relation on A*. This is simply a specific set R of ordered pairs (x, y) where $x, y \in A$. It may be symbolized to look like a regular relation by writing xRy to mean $(x,y) \in R$. For instance, when the relation R given by

$$R = \{(x, y) \in \mathbb{N} \times \mathbb{N} \mid x < y\}$$

is written as xRy, this corresponds precisely to the familiar idea $x < y$ where the symbol R is replaced by '<'. It can be pictured visually as a relation by plotting all the points (x, y) where x and y are whole numbers and marking the subset for which $x < y$. (Figure 10.3.)

However, this picture is rarely used. Instead the relationship of order is more usually embodied by marking two whole numbers x, y on the number line where $x < y$ is represented by drawing x to the left of y. (Figure 10.4.)

[2] Akkoc & Tall (2002); Bayazit (2006).

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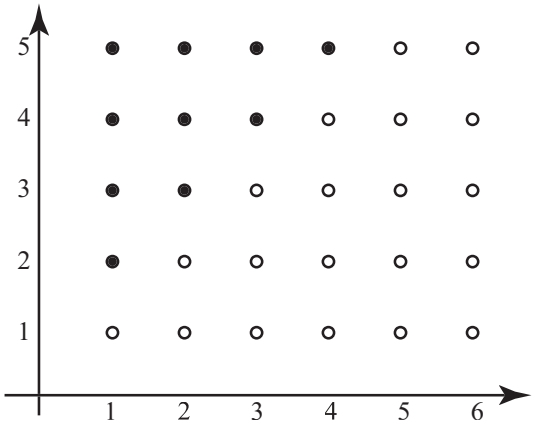

Figure 10.3. The relation $x < y$ as the subset denoted by black discs.

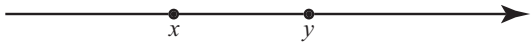

Figure 10.4. The relation $x < y$ where x is to the left of y.

The picture of a relation on A as a subset of $A \times A$ is intended to represent the full set of ordered pairs. The alternate representation of pairs in the original set A is seen more in terms of two generic elements that can be placed anywhere in A and may be imagined moving around to represent different pairs. Some relations, such as functions, are usually represented as subsets of the Cartesian product (as in Figure 10.3). Others, such as order or equivalence, are usually represented in terms of the relation between elements of the set itself (as in Figure 10.4). This can lead to problematic meanings, as we will see later in the case of equivalence relation.

2.3 *Order Relations*

An order relation is inspired by the idea of placing the elements of a set in a given order. The notion of order in number systems given in the previous chapter can be modified to apply to any set.

An order relation on a set A is a set R of ordered pairs (x, y), where we write $x < y$ if the ordered pair $(x, y) \in R$, which satisfies:

(O1) Given $x, y \in A$, then *precisely one* of the following holds:

$$x < y \quad \text{or} \quad y < x \quad \text{or} \quad x = y,$$

(O2) If $x < y$ and $y < z$ then $x < z$.

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Axiom (O1) is called *the law of trichotomy*. Only one of the three relations can hold at once. It implies that if $x \neq y$ then either $x < y$ or $y < x$ but not both. As examples, we have the natural numbers $\mathbb{N}$ or the real numbers $\mathbb{R}$ with the usual order that allows them to be embodied by visualizing them on a line from left to right so that if $x < y$ then x is to the left of y.

Likewise we can put the twenty-six letters of the alphabet in alphabetical order, $A < B < \cdots < Z$. This can be extended to the alphabetical order of words where if the first letter of one word comes before the first letter of the second, then the first word is placed before the second, but if first letters are the same, the order is determined in the same way by looking at the second letter, then the third and so on. For example, we would have $AAB < AB < B < BC < BDA < \cdots < ZZZZ$.

However, the order need not be alphabetic. An order relation on a set of words simply requires them to be placed in a specific sequence, and there is no restriction on how this is done. For instance, we may choose to order the three words *CAT*, *SAT* and *MAT* as $MAT < CAT < SAT$.

The notion of order relation is far more general than anything a student will have met before; for instance, one can take the set of natural numbers $\mathbb{N}$ and an extra element ω that is different from them all, then define an order on the extended set consisting of the elements of $\mathbb{N}$ and the element ω by using the standard order on $\mathbb{N}$ and defining $n < \omega$ for all natural numbers n. In this sense we can embody a context in which the usual set of natural numbers potentially continues forever, though nevertheless there is an additional element that satisfies $n < \omega$ for all of them.

This is a new and destabilizing idea, blending together the familiar order of whole numbers with an extra element that is 'bigger than them all'. This conflicts with familiar met-befores that the whole numbers grow 'without limit' and will be problematic for a learner who cannot imagine that anything can be 'bigger than all the natural numbers.' The element ω is not 'bigger' than all the natural numbers. It simply satisfies the formal order $n < \omega$ in the set consisting of $\mathbb{N}$ and the additional element ω. Nevertheless, the human brain, rich with connections built up over the years, naturally uses these links, consciously or unconsciously, and it requires a great effort of will to attempt to focus only on links built up through formal proof from definitions.

2.4 *Equivalence Relations*

The concept of equivalence occurs throughout embodiment and symbolism. For instance, there is the equivalence of fractions, algebraic expressions,

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free vectors, and so on, in arithmetic and algebra. There are also forms of equivalence in Euclidean geometry in terms of congruence of triangles and parallel lines.

Formally, an *equivalence relation* ~ on a set S is defined to satisfy the axioms:

(E1) $x \sim x$ for all $x \in S$,
(E2) If $x \sim y$ then $y \sim x$,
(E3) If $x \sim y$ and $y \sim z$ then $x \sim z$.

Notice that (E3) is precisely the same as (O2); however, the two identical looking axioms operate in different contexts and behave differently.

For instance, consider the following axiom:

(E3)′ If $x \sim z$ and $y \sim z$ then $x \sim y$.

In the context of the axioms (E1) and (E2), the axiom (E3)′ can be used to establish (E3) as follows:

Given the left-hand side of (E3): $x \sim y$ and $y \sim z$. By (E2) this gives $x \sim y$ and $z \sim y$, and then (E3)′ implies $x \sim z$.

The converse follows by a similar argument, so in the context of an equivalence relation, (E3) and (E3)′ can be interchanged and either can be used as part of the definition.

However, in the context of order, the two axioms are not equivalent. Given the order relation $m < n$ on the set of whole numbers, axiom (E3) is satisfied but (E3)′ is not. From $2 < 5$ and $5 < 9$ we may deduce that $2 < 9$, but the inequalities $3 < 5$ and $2 < 5$ do not imply $3 < 2$.

This leads to a surprising principle, which is important to make explicit:

The contextual role of an axiom or definition: A single axiom or definition may operate in different ways in different contexts and a formal structure depends not only on the roles of individual axioms and definitions but also on the interplay between them all within the specific context.

For instance, the notion of a relation on $\mathbb{R}$ is defined to be a subset of $\mathbb{R}^2$. Yet when Abe Chin asked the following question in a highly rated mathematics department[3], most students either did not respond or said that they did not understand the question:

[3] Chin (2002); Chin & Tall (2001, 2002).

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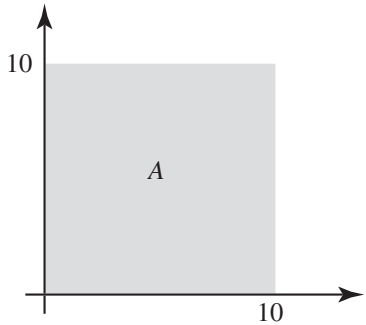

Figure 10.5. The relation A as a subset of $\mathbb{R}^2$.

$A = \{(x, y) \in \mathbb{R}^2 \mid 0 < x < 10, 0 < y < 10\}$. Is A an equivalence relation on $\mathbb{R}$?

The subset A is the interior of a square in the plane. (Figure 10.5.) It is not an equivalence relation because it fails to satisfy (E1). For instance, $11 \in \mathbb{R}$ but $(11, 11) \notin A$.

Out of 15 first-year and 15 second-year mathematics students interviewed, *no* first-year student and just one second-year student gave a correct response (using the counter-example $(11, 11) \notin A$). Of the others, those who wrote comments were all able to write down the three axioms, but often said that they did not understand the question and none mentioned the necessity that (E1) must apply to *all* elements in the set.

On closer investigation, it became apparent that the students thought of an equivalence relationship not as a subset of $A \times A$ as in Figure 10.5, but as a relationship between elements in A.

This interpretation became even more apparent in a question asked of 277 students in the same university:[4]

> Let $X = \{a, b, c\}$ and the relation $\sim$ be defined where $a \sim b$, $b \sim a$, $a \sim a$, $b \sim b$, but no other relations hold. Is this an **equivalence relation**? If not, why?

Again, this is not an equivalence relation because $c \sim c$ is not specified, even though $c \in X$, violating (E1). However, only 139 students (50%) gave a correct response with a correct supporting reason. Sixty-eight others (25%) gave the correct answer but offered an incorrect reason, claiming that the relation is *not transitive*.

[4] Chin & Tall (2002).

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The relation is certainly transitive, because whenever $x \sim y$ and $y \sim z$ are satisfied then so is $x \sim z$. Because the relationship is specified only between the two elements a and b, any application of the transitive law must be trivially true. For instance, given $a \sim a$ and $a \sim b$, then putting $x = a$, $y = a$, $z = b$ gives $x \sim z$, which translates to $a \sim b$, which is true.

The problematic met-before here is that in many situations the transitive law implicitly assumes that x, y, z are *different*. For instance, the order relation $x < y$ automatically implies x and y are different and is embodied by points on a line where x is to the left of y.

In examples that the students have met before, the relation concerned may require the elements involved to be different. For instance, if the notion of two lines in a plane being parallel is defined by the fact that they do not meet, then a line cannot be parallel to itself because it meets itself all along its whole length. In this case, (E1) fails but (E2) is true. Meanwhile (E3) is true in the case that a, b, c are all different, but it fails if a and c are the same and a is parallel to b (because then a is parallel to b and b is parallel to a but a is not parallel to a).

The Greeks treated the notion of congruence as a relation between two *different* triangles, so they did not allow a triangle to be congruent to itself, even though today we would be happy to allow this as part of a modern theory to allow congruence to be an equivalence relation.[5]

The same occurs with relations in everyday life. I have a brother Graham. He is my brother and I am his brother, but only a mathematician might say that I am my own brother.

When we make statements in everyday arithmetic, we use the information to say precisely what we know, for instance, we say that '3 + 2 is equal to 5' and '3 + 2 is less than 6', but rarely that '3 + 2 is *less than or equal* to 6' as we already *know* that it is not equal. With such a background of experience, our interpretation of the notion of an equivalence relation can involve subtle met-befores that are not explicitly given in the set-theoretic definition. In particular, the symmetric law $a \sim a$ on the set S must apply to *all* $a \in S$ and the other two laws must include the possibility that the elements involved need not be different.

These examples reveal the rich connections built up through previous experience that may cause problematic met-befores in new situations that students face in their transition to formal mathematical thinking.

[5] Amazingly, the notion of 'congruence' is not given a name in the theory of Euclid. It is a process to go through to show that two (different) triangles that have certain corresponding properties, such as having all three corresponding sides equal, would then have all other attributes equal.

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3. Real Numbers and Limits

The shift from rational numbers to real numbers proves to be a major watershed for many students. In school, students meet irrational numbers such as $\sqrt{2}$, π and e, and begin to realize that the number line has numbers on it that are not rational, though it is not clear precisely what these irrational numbers are.

Historically the problematic nature of these new numbers is reflected in the name *irrational*, meaning they are not rational numbers calculated by dividing one whole number by another. John Monaghan found that some students aged sixteen to eighteen sensed that these additional numbers were somehow 'improper'.[6] They could not be produced by a simple calculation with whole numbers. They had infinite decimal expansions, such as $\sqrt{2} = 1.414\ldots$ and these were often considered as 'infinite' numbers, not because they were infinite in size, but because they were infinite *in extent*. They could never be calculated accurately because the decimals go on forever.

A particular case is the repeating decimal $0.999\ldots$. This can be approximated by working it out to, say, n decimal places, which equals $1 - (1/10^n)$. This approximation can get as close as one likes to 1, *but it is never equal to 1*. The result is that the vast majority of students, and many teachers, believe that $0.999\ldots$ is not equal to 1; it is *just less*.[7]

Given the widespread occurrence of this interpretation, there must be an underlying reason for it. The biological brain must deal with a term $s_n = 1 - (1/10^n)$, and consider what happens 'as n increases'. It may be hypothesized that the brain makes sense of each term in the same way, so it imagines a single variable term that changes and increases as n increases. This is a compression of knowledge from a sequence of different terms to a single varying term. This variable term can get as close as we like to 1 but is never equal to it, so it is natural to conceive the limiting case as a quantity that is arbitrarily close to, but never equal to 1.

The idea that quantities can be 'arbitrarily close' or 'arbitrarily small' is a phenomenon that leads naturally to a mental concept that is arbitrarily small, but not zero. Such a notion arises both in embodiment and in symbolism.

This idea of being arbitrarily small, but not zero, is consistent with our experience of points marked with a pencil or lines drawn on paper.

[6] Monaghan (1986).
[7] Cornu (1991).

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A point drawn in this way is small, but not so small that it cannot be seen, so it may be imagined as being 'as small as possible'. A line, though thin, has a physical width and, in the quest for precision, it is drawn practically as thin as possible. But it is not drawn without any thickness at all. As a result, our imagination of points and lines based on perception is as small as possible and as thin as possible. Even though we may propose the Platonic idea of a point having position and no size and a line having length but no breadth, this must operate in the same biological brain with the practical met-befores of points and lines that *do* have a size, which are as small as possible.

This is evident in the history of calculus, where Leibniz formulated his 'principle of continuity' that said:

> In any supposed transition, ending in any terminus, it is permissible to institute a general reasoning, in which the final terminus may also be included.[8]

In a supposed transition with a sequence of terms such as $\frac{1}{2}, \frac{1}{4}, \ldots, (\frac{1}{2})^n, \ldots$ which get smaller and smaller but are always positive, Leibniz's principle suggests that the 'final terminus' is a quantity that is arbitrarily small but positive. An infinitesimal is a natural product of the functioning of the human brain.

In the same way, an infinite decimal expansion such as $1/3 = 0.333\ldots$ or $\sqrt{2} = 1.4142\ldots$ is often imagined initially as *approaching* the limit, rather than *being* the limit.

In 1986[9], I used the principle of continuity of Leibniz to describe the notion of *generic limit*, which need not be a limit in the mathematical sense but 'is the concept of limit that the individual holds in his or her mind as a result of extrapolating the common properties of the terms of the sequence.' This idea was also extended to various other related concepts such as a *generic tangent*[10], which may not be a true tangent, but is drawn as a line that 'touches the curve at a single point and does not cross it.' Generic ideas arise from considering specific examples as typifying a more general concept.

A telling example of this phenomenon occurred in the work of one of my research students, Lan Li, who asked undergraduate students studying for a degree in teaching mathematics:

[8] Leibniz, translated in Child (1920), p. 147.
[9] Tall (1986a, b).
[10] Tall (1986b), p. 74.

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(A) Can you add 0·1 + 0·01 + 0·001 + ... and go on forever and get an exact answer? (Y/N)

(B) $1/9 = 0\cdot\dot{1}$. Is 1/9 equal to $0\cdot1+0\cdot01+0\cdot001+\ldots$? (Y/N)

The favoured response was *No* to (A) and *Yes* to (B).[11] Interviewing the students at the time, it became evident that the symbolism was being read from the left to the right so that $0\cdot1+0\cdot01+\ldots$ represents a potentially infinite process that can never be completed but the second shows how $1/9 = 0\cdot1+0\cdot01+\ldots$ can be divided to get as many terms as desired.

During this course, I did my very best to give the students experience of programming convergent sequences, to see that some converged faster than others, and I carefully explained that an infinite decimal is a notation for the *fixed quantity* to which the sequence of finite approximations can get as close as is practically required. I gave examples to illustrate the principle. In particular, I showed that 0.999…9 to n decimal places is equal to $1-(1/10^n)$ and that, for any given positive value of ε, we can find an N such that if $n > N$, then 0.999…9 to n decimal places differs from 1 by less than ε. I explained that N can be found by writing $1/\varepsilon$ as a decimal and choosing N such that $10^N > 1/\varepsilon$; then, whenever $n > N$, we have $1/10^n < \varepsilon$ and so the difference between 1 and 0.999 … 9 to n decimal places is less than ε as required. I focused on the fact that the limit was the *fixed number* that the sequence tended to, according to the definition.

Before the class, most of the students (21 out of 25) believed that 0.999… is less than 1, and I felt confident that I could change their views by giving a coherent explanation. Two weeks later I repeated the question and 21 out of the 23 present at the time still replied that 0.999… is less than 1. In the discussion that followed, the main argument put forward by the students was that they were sure that '0.999 repeating' never quite reached 1, so it was not proper to attempt to *define* it to be otherwise.

When Lan Li, the research student working with me, wrote up the experiment for her master's degree, she said that it was not proper for her to write that her supervisor failed to explain the idea satisfactorily. My response was that if this had happened when I was trying my best to give a good explanation, then there must be a serious underlying reason.

As I reflected on this and other evidence, I recalled another study that amazed me at the time, but made more sense upon reflection. Nicholas

[11] See Li and Tall (1993). The response *Yes* to (a) and *No* to (b) was given by 18 out of 25 students (72%) before the course and 14 out of 23 (61%) after.

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Wood questioned pure mathematics students who had received more than a year's training in analysis techniques. When he asked 'Is there a *least* positive real number?' and 'Is there a *first* positive real number?' a sizeable minority said there was a first, but not a least.[12]

There are underlying experiences that can give rise to these conflicting responses. Geometrically and algebraically there cannot be a least positive real number x, because $\frac{1}{2}x$ is still positive and smaller. However, in the symbolism of arithmetic, writing a number to, say, four decimal places, there *is* a first nonzero number, namely 0.0001 so, perhaps this is extrapolated to the case of an infinite number of places in the form 'zero point zero, zero, zero, repeating, with a one at the end.' This seems to relate to the difference 1 – 0.999…, which may be imagined as a string of zeros with a one in the 'last' decimal place.

Lakoff and Núñez expressed the general idea in terms of what they termed as the 'basic metaphor of infinity':

> We hypothesize that all cases of infinity – infinite sets, points at infinity, limits of infinite series, infinite intersections, least upper bounds – are special cases of a single conceptual metaphor in which processes that go on indefinitely are conceptualized as having an end and an ultimate result.[13]

Formulated in these terms, this is essentially a restatement of the principle of continuity of Leibniz, although the authors formulate it in a new way. For instance, they explain the notion of infinite decimal in terms of a succession of finite decimal representations $R_1, R_2, \ldots$ where R_n is the set of real numbers that have decimal representations with n decimal places. The Basic Metaphor of Infinity is invoked to declare that the system has an end and an ultimate result that is the set R_∞ of infinite decimals. Lakoff and Núñez explain that the elements of R_∞ are not the real numbers but the numerals that are *names* for the real numbers and their precise meaning requires further applications of the Basic Metaphor.

At this point, what they do not say is that – for the student and for the vast majority of ordinary mortals – infinite decimals are slightly strange (improper) numbers that are infinite in extent but implicitly share the properties experienced with finite decimals.

In the finite case the different numerals in R_n are all different decimals. To the biological brain using the same neural circuits to interpret these

[12] Wood (1992).
[13] Lakoff & Núñez (2000), p. 158.

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improper numbers, different infinite decimals also have *different* values. The finite sequence 0.999…9 (to n decimal places) is less than 1.000 … 0, *so the improper number 0.999… is less than 1*!

The evidence lies not only in the many responses from students that support this view, but also in the amazing result of Wood that a substantial number of undergraduates believe that there is not a least positive number, but there *is* a *first* positive number, which in R_n is 0.000 … 01 and leads to the belief that the first positive real number in R_∞ is 'zero point infinite zeros, one' that can be written as 1 – (0.999 …).

This illustrates the need to complement a top-down cognitive science 'idea analysis' with a bottom-up cognitive development analysis of how students make sense of new ideas based on their previous experience. In *Where Mathematics Comes From* by Lakoff and Núñez, and *The Way We Think* by Fauconnier and Turner, the analyses of conceptual blends are performed using high-level concepts that are available in various ways to modern philosophers, mathematicians and cognitive scientists. Such concepts are not usually available to individuals attempting to make sense of new ideas for the first time. For instance, Gilles Fauconnier and Mark Turner interpret the complex numbers as an emergent blend using the modern axiomatic notion of field, which was certainly not available to the individuals earlier in history. What is more important, both in history and in the development of students, is to try to understand how individuals may make sense of a system not just in terms of the viewpoint of an expert, but *from the viewpoint of the learner*, or more precisely, in terms of the learner's personal met-befores.

In attempting to describe how students understand real numbers and limits, it is essential to see where the students are coming from. They come from experiences with what I shall call 'good-enough' arithmetic, using calculations to an appropriate accuracy to get an answer that is good enough for the purpose in hand. Good-enough arithmetic concerns the practical mathematics of the world that we live in, where straight lines are drawn with physical implements and quantities are given to an appropriate accuracy. Sometimes it is good enough for π to be 3.142 or $^{22}/_{7}$ and, in a given situation, either of these may be satisfactory whereas in other situations a more accurate value may be required. For a carpenter or an engineer, the practical mathematics of good-enough arithmetic is good enough.

As learners grow in sophistication, they learn the strengths and weaknesses of good-enough ways of operating with numbers. They know that if they cut a quantity in half, then in half again and so on, theoretically this

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can go on forever, though in a very practical sense the quantity is soon so small that it cannot be further divided.

Good-enough arithmetic works in everyday calculations, but it is not perfect. Floating point arithmetic on computers gives sufficiently accurate answers for most practical purposes, but it does not satisfy all the rules of arithmetic. For instance, $(a+b)-a$ may not equal b in a practical calculation when $a=1$ and $b=10^{-1000}$ because $a+b$ is rounded to 1 as a floating-point number with a limited number of places and the result is $(a+b)-a=1-1=0$, even though $b \neq 0$. More seriously, calculating ratios of small numbers can give huge errors, and good-enough arithmetic becomes problematic for calculating numerical derivatives in the calculus.

Expert mathematicians, seeking perfection, move from good-enough arithmetic to the theoretical mathematics of real numbers to obtain perfect arithmetic. At a higher level, this leads to the formal limit concept. However, many practical professionals in engineering and other occupations essential for our society do not require the formal notion of limit, only a firm grasp of good-enough arithmetic and the ability to model problems and calculate solutions to the required accuracy.

This illustrates the difference between a *theoretical* limit related to the need to think coherently with good-enough arithmetic in applications and the *formal* limit as given by the formal definition in mathematical analysis.

Pure mathematicians see the formal limit concept as a watershed. Students who can operate with it in a formal way are believed to have the potential to become genuine mathematical thinkers to formulate proofs in axiomatic formal mathematics. The reality is somewhat different. Students meeting the concept in their first course of real analysis must build on their current knowledge structures. They may do this in qualitatively different ways from the formal mode of thinking seen as the ultimate form of expression by expert pure mathematicians.

Applied mathematicians are content with a theoretical approach that uses pure mathematics to formulate mathematical models in applications to solve and predict the possible outcomes of a given situation.

3.1 *Mathematical Constructions of the Real Numbers*

In the latter part of the nineteenth century, the concept of real number was given a formal construction in different ways by Georg Cantor[14] and Richard Dedekind[15].

[14] Cantor (1872).
[15] Dedekind (1872).

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Cantor's method began with a particular type of sequence, $s_1, s_2, \ldots, s_n, \ldots$, of rational numbers introduced by Cauchy, where the terms of a sequence 'get close to each other' in the following sense:

> **Definition:** A Cauchy sequence is a sequence $s_1, s_2, \ldots, s_n, \ldots$ of numbers so that, given any rational $\varepsilon > 0$, a value of N can be found so that when $m, n > N$, the difference between s_m and s_n is less than ε.

Given two Cauchy sequences $a_1, a_2, \ldots$ and $b_1, b_2, \ldots$ then we can consider whether the sequence $a_1 - b_1, a_2 - b_3, \ldots$ tends to zero. If this happens then we can say that two such Cauchy sequences are 'equivalent'.

This idea essentially allows us to say that equivalent Cauchy sequences have the same limit without actually needing to calculate the limit itself. Cantor's idea was to *define* a real number to be given by a Cauchy sequence of rational numbers, where two equivalent sequences defined the same real number. A rational number r gives rise to a constant Cauchy sequence where every term is equal to r. The rational numbers can then be regarded as special cases of real numbers[16] and are now a subset of the extended system of real numbers.

Dedekind's method, inspired by Cantor, envisaged the rational numbers on the line being cut into two disjoint sets, L and U, in such a way that every rational in L is smaller than every rational in U. This separates all the rationals into a lower set L and an upper set U, something like Figure 10.6.

Figure 10.6. A cut on the rational number line into sets L (grey) and U (black).

This picture represents only *rational* numbers and leads to essentially two distinct forms of cut. One occurs at a rational number r, where L is the set of rational numbers less than r; U is the set of rationals greater than r; and r itself is included either in L or in R, which will be here chosen in L for definiteness. The other does *not* occur at a rational number, such as happens when U is the set of all positive rationals with square greater than 2, and L contains the rest. Essentially this cut corresponds to a new type of number that is *not* rational, namely $\sqrt{2}$.

Dedekind defined an addition for cuts by adding the elements in the two lower sets together and those in the two upper sets together to give

[16] A subtle use of the idea that the rational numbers and their arithmetic are the same crystalline concept as the equivalence classes of constant sequences.

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the sum as a new cut. He defined subtraction, multiplication and division of cuts, which needed a little more ingenuity to cope with negative values. These 'cuts' were now considered as 'numbers', some corresponding to the rational numbers where the cut occurs and some new cuts that are the *irrational* numbers. Dedekind's methods effectively expanded the number line to include both rationals and irrationals. In a very 'real' sense, both Cantor and Dedekind 'completed' the number line by introducing the irrationals.

Once the number line is seen to be 'complete', with both rationals and irrationals, the scene is set for a rigorous approach to the calculus in terms of the theory of mathematical analysis.

The completion was later christened 'the Cantor-Dedekind Axiom', stating that

> The real numbers are order-isomorphic to the geometric line.[17]

This essentially fulfils the vision of Descartes, who saw points in the plane expressed as pairs (x, y) of numbers and, in particular, points on the x-axis in a direct ordered correspondence with the number line. The real number line is now seen as a blend of geometric points and symbolic decimals.

The real number system $\mathbb{R}$ satisfies the axioms for an ordered field, together with the completeness axiom, which states

(C) Any Cauchy sequence in $\mathbb{R}$ tends to a specific limit in $\mathbb{R}$.

For instance, the sequence of approximations, $a_1 = 1.4$, $a_2 = 1.41$, $a_3 = 1.414, \ldots$, where a_n is the value of $\sqrt{2}$ to n decimal places, is a Cauchy sequence (because for $m, n > N$, the terms a_m and a_n agree at least for N decimal places, so they differ by at most $1/10^N$).

The completeness axiom (C) essentially guarantees that any infinite decimal has a numerical value. Later we will show that any ordered field satisfying the axiom of completeness has the same arithmetic structure as the arithmetic of infinite decimals and that it has a unique crystalline structure that fits together in such exquisite ways that mathematicians imagine it as a unique Platonic object.

To be able to build this new level of mathematical sophistication, we must now introduce the formal notion of limit.

[17] See, for instance, The Cauchy-Dedekind Axiom in Wikipedia, retrieved from http://en.wikipedia.org/wiki/Cantor-Dedekind_axiom (Accessed July 28, 2012).

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3.2 *Introducing the Limit Concept*

The formal definition of the limit concept states that a sequence $a_1, a_2, \ldots, a_n, \ldots$ tends to a limit a (which is fixed) if the following condition is satisfied:

Given any $\varepsilon > 0$, there exists a whole number N such that
$n > N$ implies $|a_n - a| < \varepsilon$.

Making sense of this definition is problematic, not least because it has several quantifiers. Written in symbolic language using the quantifiers ∀ (for all) and ∃ (there exists), it becomes

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \quad \text{such that } \forall n > N, |a_n - a| < \varepsilon.$$

This involves three quantifiers nested within each other, alternating 'for all' and 'there exists' in a manner that can be complicated to handle. Furthermore, the constructions of the real numbers formulated by Cantor and Dedekind are based on visual imagery and experience of operating with numbers. The formalism of Hilbert is intended to be based purely on set-theoretic deductions in terms of a system satisfying the axioms of a complete ordered field.

We begin our exploration of student conceptions of the limit concept based on a longitudinal study in which Marcia Pinto interviewed eleven students on seven occasions at intervals throughout a twenty-week analysis course[18], seeking underlying themes using grounded theory.[19] The grounded theory data led to categories that distinguished whether the student constructed the concept of limit by *giving meaning* to the definition based on the student's concept image or *extracting meaning* from the definition by operating with the quantified definitions to build formal proof. These two approaches are related to *a natural approach* building on the concept image and *a formal approach* building formal theorems based on the formal definition.

The concept image as defined in Tall & Vinner 'includes all the mental pictures and associated properties and processes.'[20] This underlines the need to include not only pictures, but also symbolism, as encountered in practical and theoretical mathematics. This is consistent with the historical

[18] Pinto (1998); Pinto & Tall (1999, 2001, 2002).
[19] Strauss & Corbin (1990).
[20] Tall & Vinner (1981), p. 152.

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use of the term 'natural' in 'natural philosophy' including geometry, arithmetic, algebra, and calculus to describe natural phenomena prior to the introduction of formal mathematics at the end of the nineteenth century. This leads to a categorization of the development of proof into:

- A natural approach based on theoretical mathematics involving embodiment, or symbolism, or a blending of the two;
- A formal approach using the formal mathematics of set-theoretic definition and deduction.

We begin by considering the data from four selected students.[21]

4.1 *A Natural Approach Blending Embodiment and Symbolism*

Student Chris typified an approach building from embodiment and symbolism to construct the formal definition. To write down the definition of convergence, he drew a picture and imagined the dynamic process as the

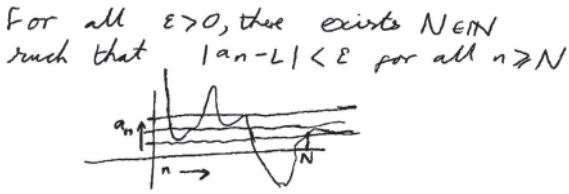

Figure 10.7. Chris's conception of the limit concept.

sequence (a_n) of terms tended to the limit L. (Figure 10.7.)

As he drew the diagram, he motioned with his hands to suggest the ideas underlying the definition, saying:

> I don't memorize that [the definition of limit]. I think of this [the picture] every time I work it out, and then you just get used to it. I can nearly write that straight down.
>
> I think of it graphically … you got a graph there and the function there, and I think that it's got the limit there … and then ε once like that, and you can draw along and then all the … points after N are inside of those bounds.… When I first thought of this, it was hard to understand, so I thought of it like that's the n going across there and that's a_n.… Err, this shouldn't really be a graph, it should be points.

[21] The material in the following section is based on the doctoral research of Marcia Pinto (1988).

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In his final remark, he recognized that he had drawn a continuous graph with a motion of his pencil, instead of a sequence of points. However, the thrust of his dynamic argument remained valid and allowed him to build up the formal definition from his embodied imagery.

Throughout the whole course he seemed to be negotiating with the ideas. For instance, he believed that a formal proof must include *all* the logical steps to build from the given assumptions. He did not allow the quotation of results previously established because he was adamant that each 'proof' must be complete in itself. It was not until the eighth week of the twenty-week course that he gave up this ideal when his proofs became interminably long.

He also played with various possible alternatives for definitions. For instance, instead of starting with ε and looking for a value of N, he considered starting with N to find ε, before deciding that the standard definition is more appropriate.

He enjoyed the tension of the challenge. He had clearly faced mathematical challenges before and felt the thrill of success. Now he was maintained on an emotional high level that seemed to give him pleasure even when under stress. He always looked for clarity and precision in his ideas, and surmounted errors by refocusing his attack.

Chris's approach developed a rich blend of embodiment, symbolism and formalism where formal ideas were supported by embodied imagery and fluent symbolic operations. His approach is 'natural'. It 'gives meaning' to the definition focusing on dynamic visual imagery and blending this with meaningful manipulation of quantified symbolic definitions.

4.2 *A Formal Approach Based on Definition and Proof*

Student Ross coped with the definition by repetition:

> Just memorizing it, well it's mostly that we have written it down quite a few times in lectures and then whenever I do a question I try to write down the definition and just by writing it down over and over again it gets imprinted and then I remember it.

When he investigated the limit of a sequence, he used the definition. He declared that the constant sequence 1, 1, 1, … tends to 1, because, given any $\varepsilon > 0$, take $N = 1$ and for $n > 1$ we have $|a_n - 1| = 0$, so clearly $|a_n - 1| < \varepsilon$ as desired.

Many other students, however, have the linguistic met-before that the phrase 'tends to' means 'gets close to, but is not equal to the limit.' This is

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emphasized by many examples such as $1/n$, which *tends* to 0, but the term $1/n$ is never actually zero itself.

Many students consider a constant sequence as a special case that feels different from a sequence that tends to a limit, getting ever closer to the limit, which it never reaches. Ross had a much more subtle insight. He thought very carefully about the idea of convergence and realized that for a certain value of ε, some sequences would require much larger values of N than others, and so one could talk about some sequences converging more slowly than others. In a later tutorial, he brought this topic up and declared that the constant sequence is 'the fastest converger of all'. This is the mark of a true mathematician. While other students might include the constant sequence as a separate case, Ross considered it as a central example.

Ross drew a picture in interview (Figure 10.8), but insisted that he built his ideas by thinking about what was happening dynamically. In an interview, he explained:

> Well, before … I saw anyone draw that, it was just umm … thinking basically as n gets larger than N, a_n is going to get closer to L, so that the difference between them is going to come very small and basically, whatever value you try to make it smaller than, if you go far enough out then the gap between them is going to be smaller. That's what I thought before seeing the diagrams … something like that.

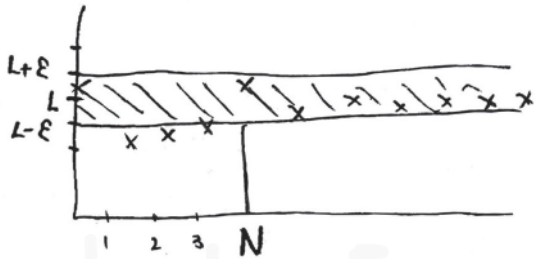

Figure 10.8. Ross's sketch of convergence.

In his responses he used dynamic metaphorical language in which 'n gets larger than N,' 'a_n is going to get closer to L,' 'the difference between them is going to come small,' 'the gap between them is going to be smaller.' Thus, although he did not build his ideas from a conceptual embodiment in a picture viewed dynamically, he used functional embodiment of the dynamic changes in the process of tending to a limit. In doing so, he built on the formal definition and suppressed his concerns over the problematic nature of his visual interpretation of the limit concept.

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Over time he became highly proficient in manipulating the logical symbols, such as using the quantifiers ∀ (for all) and ∃ (there exists). He realized that to show a 'for all' statement is false needs just one counter-example and using the symbol ¬ for 'not' leads to the simple rule that one can replace ¬∀ by ∃¬. In the same way, to show that a 'there exists' statement is false requires the proof that the statement is false in all cases, so ¬∃ can be replaced by ∀¬. He used these principles to deduce what is meant for a sequence not to be convergent by writing

$$\neg\left(\forall \varepsilon > 0\, \exists N\, \forall n > N : \left|a_n - L\right| < \varepsilon\right)$$

and moving the negation sign successively over the quantifiers, swapping them around to get

$$\exists \varepsilon > 0\, \forall N\, \exists n > N : \neg \left|a_n - L\right| < \varepsilon$$

so

$$\exists \varepsilon > 0\, \forall N\, \exists n > N : \left|a_n - L\right| \geq \varepsilon$$

Ross is a 'formal' learner, building a rich knowledge structure for the thinkable concept of limit by following through the proofs and getting a sense of the meaning of the logical structure. He uses logical symbolism to represent the definition in a compact form and manipulates it easily and logically while also thinking about the limit process as a functional embodiment as the terms get as close as desired to the limiting value.

4.3 *A Problematic Embodied Route*

Colin based his idea of convergence on the embodied movement along a descending curve that he drew and described in dynamic terms (Figure 10.9), saying:

> … um, [I] sort of imagine the curve just coming down like this and dipping below this point which is ε … and this would be N. So as soon as they dip below this point then … the terms bigger than this [pointing from N to the right] tend to a certain limit, if you make this small enough [pointing to the value of ε].

He was not able to formulate this idea in a way that enabled him to produce a formal proof. For example, when asked to prove the following

If $a_n \to 1$, prove that there exists $n \in \mathbb{N}$ such that $a_n > \frac{3}{4}$ for all $n > N$,

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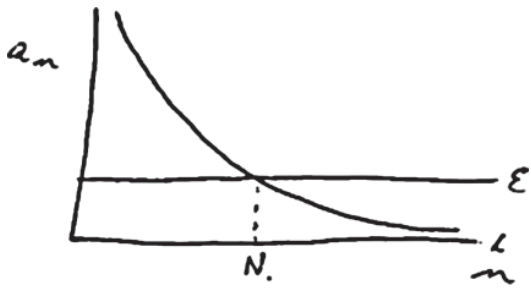

Figure 10.9. Colin's mental image of convergence.

he said:

> It seemed to be a silly question that … if a_n tends to 1 then if you question when a_n is greater than $\frac{3}{4}$ … this is a bound, it seems … I don't know why.

It seemed self-evident to him that if the terms get closer to 1, eventually they must be bigger than $\frac{3}{4}$. However, there is not a specific formula given for a_n or a numerical value for ε, so it is not possible to calculate a numeric value for N.

As the course progressed, he sensed the conflict between his embodied imagery and the formalism. For instance, his dynamic imagery suggested that 0.999… (zero point nine repeating) gets 'closer and closer' to 1 without reaching it, but the formal definition tells him that the limit *is* 1. Towards the end of the course, the problematic aspects of his conception of limit remained, as he commented:

> It's sort of … I understand it should be 1 … and that the limit of the sequence is actually 1 just … It's down to notation. It just it's a bit hard to let go of 0.9999 recurring …

4.4 *A Problematic Symbolic Route That Is Essentially Procedural*

Student Rolf built on his symbolic experience with algebra and arithmetic to interpret the definition of convergence as a calculation. For him the definition said:

> specify the formula for the nth term of the sequence and how small you require ε and I will be able to compute the value of N so that when $n > N$, then the nth term differs from the limit by less than ε.

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He was able to use this computational procedural approach to deal with specific cases where he was given an explicit formula for the term and a numerical value of ε to work with. For him the procedure was operational in specific cases, but it was not meaningful if he did not have an explicit formula and a numerical epsilon to use in a computation. Over time, he explained that he was beginning to become accustomed to the definition without really understanding what is going on.

> Err … I (once?) thought the definition was stupid. I once thought there was no point in having … complicated definitions. But now I think it's okay. I mean, *I am getting used to it*, but I thought the definition was, err, I didn't really fully understand the definition.

However, he could not understand what it meant to say that a sequence does *not* tend to a limit L:

> … it's a pretty useless definition, isn't it? Do you see what I'm saying? Because if it doesn't tend to a limit, then you then … what you are going to imply for L? I think that's the definition, but it's pretty useless.

Rolf was not comfortable studying analysis. He sensed how limits worked dynamically and could verify the limit definition by a calculation if he had the necessary information, but he did not grasp how to reason with the formal definition in formal proofs. After a term he changed his course to read applied mathematics.

4.5 *Classifying Routes to Formal Mathematical Thinking*

The four examples in the preceding section are carefully selected from eleven student responses to represent a spectrum of individual conceptions of the limit. Broadly speaking, they can be seen as conceptions that have a range of differing supportive and problematic aspects.

Chris built by extracting meaning from visual thought experiments and blended his embodiment with quantified symbolism to succeed in developing formal proof. In general, for a natural learner building from embodiment, it is essential to translate embodied reasoning into quantified symbolism.

Ross took a fundamentally formal approach, extracting meaning from the definition by repeating it and working through the proofs until he had sufficient familiarity to reflect on them and make sense of the axiomatic formalism. He had a supportive logical knowledge structure, but he suppressed his imagery, which he found to be limited, even problematic.

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His focus on verbal logic enabled him to make sense of aspects that many students found to be problematic (such as the convergence of constant sequences) as coherent parts of a supportive formal knowledge structure.

Colin attempted to build his idea of a limit by extracting meaning from embodied imagery but was unable to make sense of the multiple quantifiers. He had an intuitive sense of the limit concept but various aspects of his imagery, such as a sequence 'approaching a limit', proved to be problematic in making sense of the formal definition.

Rolf was classified as 'extracting meaning' from the definition but was unable to cope with the quantifiers. He based his arguments on symbol manipulation and did not draw pictures. If the formula for the nth term was given, for a given numerical value of epsilon, he could calculate a numerical value of N, so he was building on his natural symbolic experience but did not develop into a full formal approach.

Figure 10.10 shows the full cognitive development of proof, from specific and generic examples in practical mathematics, natural embodied and symbolic proof in theoretical mathematics that may develop into formal proof based on quantified set-theoretic axioms and definitions.

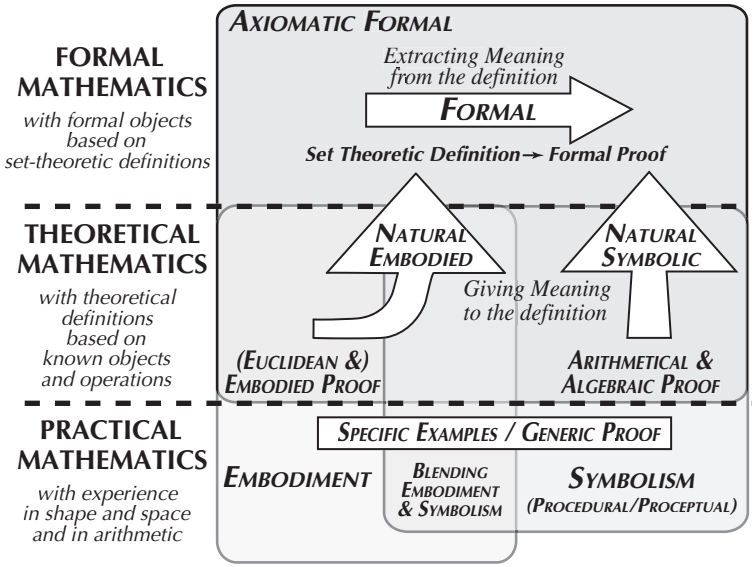

Figure 10.10. The long-term development of proof in mathematical analysis.

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5. Comparison of Theoretical Frameworks

The broad picture of the long-term development of proof has been categorized in a number of ways. Janet Duffin and Adrian Simpson used the term 'natural' to denote an approach building naturally on previous experience as opposed to an 'alien' approach in which the learner is willing to accept the definitions and work with them without needing to give them a specific meaning.[22] They also spoke of a 'conflicting' approach where the student was impeded by conflict between previous experience and new formal ideas. Pinto and Tall adopted the terminology for a 'natural' approach, but used the term 'formal' to describe the derivation of formal proof while noting that conflict could arise from both these approaches.[23]

Working at about the same time, Keith Weber saw three categories, naming them 'natural', 'formal' and 'procedural' as follows:

- a *natural* approach, giving an intuitive description and using it to lead to formal proof,
- a *formal* approach, where students had little initial intuition but could logically justify their proofs,
- a *procedural* approach, in which students learnt the proofs given them by the professor by rote without being able to provide any formal justification.[24]

In addition, he analyzed the teaching style, where the professor concerned began with an initial *logico-structural* style in which he guided the students into constructing a sequence of deductions to prove a theorem. He divided his working space on the board into two columns, with the left column to be filled in with the text of the proof and the right column as 'scratch work'. He wrote the definitions at the top of the left column and the final statement at the bottom, and then used the scratch work area to translate information across and to think about the possible deductions that would lead from the assumptions to the final result.

Interpreted in problem-solving terms, he wrote 'what is known' (the hypotheses), 'what is wanted' (the conclusion) and sought to find 'what to introduce' to build a formal argument from hypothesis to conclusion.

[22] Duffin & Simpson (1993).
[23] Pinto & Tall (1999).
[24] Weber (2004).

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Later in the course, he wrote his proofs in a more sequential *procedural* style, writing the proof down in the left column and using the right column to work out detail, such as routine manipulation of symbols. When the topic shifted emphasis to a geometric approach, he used what Weber termed a *semantic* style, by drawing pictures and introducing ideas visually, essentially building from embodiment to formal proof.

The students' responses in examinations related to the teaching style. In topics presented in a logico-structural or procedural manner, of six students interviewed, four responses were classified as formal, one as procedural and one natural. In topics taught in a semantic style, there were no formal responses, one was procedural and five were natural. Logical and structural teaching resulted in more formal responses while visual embodiments resulted in more natural responses.

Lara Alcock and Adrian Simpson used grounded theory to study two parallel analysis courses with the same syllabus, 'covering formal, definition-based approaches to convergence of sequences and series and the completeness of the real numbers'.[25] One gave standard lectures and the other used cooperative learning based on the book, *Numbers and Functions: Steps into Analysis*, by Robert Burn.[26] In the latter the students learned about analysis concepts as a natural extension of school arithmetic and algebra, working together to make calculations with sequences that tend to zero, to build formal ideas by giving meaning to quantified symbolic definitions.

Interviewing eighteen students in pairs, they found the data led to an analysis into 'visual' and 'non-visual' approaches. The visual students introduced diagrams, made gestures while explaining and preferred pictures to algebraic explanations. The non-visual students used algebraic notations, did not introduce diagrams, and were not inclined to use pictures in reasoning. Nine students were classified as visual, seven as non-visual, and two were not classified because their partner dominated the interview. In particular, when a pair included a visualizer and a non-visualizer, when the visualizer explained reasoning based on visual images, the non-visualizer usually remained silent.

The students were subdivided into three bands. In the lower band visualizers referred to limited visual information, but neither they nor non-visualizers coped with definitions or arguments.

In the middle band, the visualizers base their ideas on embodiment with some links to the definition, while they shared with non-visualizers an

[25] Alcock & Simpson (2004, 2005).
[26] Burn (1992).

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incomplete understanding of the formal concepts and relied on the book or the lecturer as the source of authority.

The higher visual band blended together embodiment and symbolism to develop a quantified formal approach. The higher non-visual band built from algebraic experience to a quantified formal approach.

In a later paper, Lara Alcock and Keith Weber collaborated to produce a two-part analysis of the development of proof categorized as 'semantic' and 'syntactic'.[27] They describe a semantic approach 'in which the prover also makes use of his or her intuitive understanding of the concepts' and a syntactic approach in which 'the prover works from a literal reading of the involved definitions'.

In the three-world framework, the visual-symbolic analysis may be considered in terms of the relationship between conceptual embodiment and operational symbolism, while the semantic/syntactic analysis relates more to the transition from embodiment and symbolism to axiomatic formalism. There may be differences in detail, but the broad development may be represented by Figure 10.11.

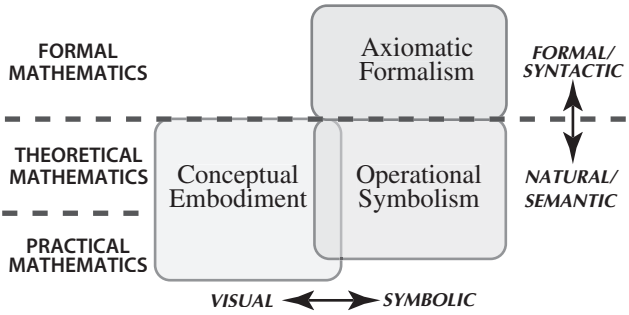

Figure 10.11. A two-by-two analysis of the growth of mathematical thinking and proof.

This framework is directly related to Bruner's enactive-iconic-symbolic modes of operation in general cognitive development. The three-world framework combines enactive and iconic as conceptual embodiment and sub-divides symbolism into the operational symbolism of arithmetic and algebra and the increasing verbal reasoning leading to the logic of axiomatic formalism. This provides a two-way analysis between the visual and symbolic in school mathematics where the visual is verbalized and the symbolism is generalized to translate into the quantified set-theoretic logic of

[27] Alcock & Weber (2004).

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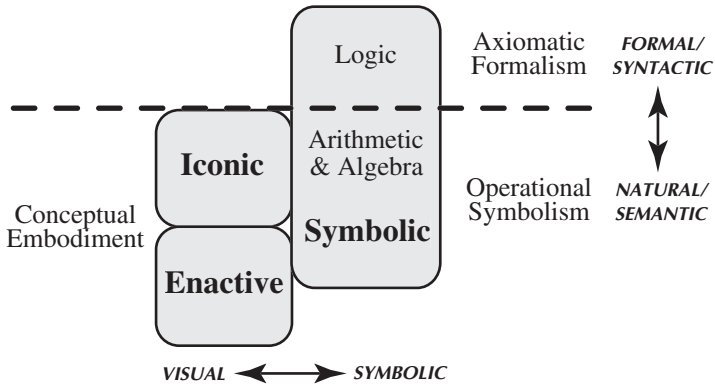

Figure 10.12. Bruner's three modes and the three worlds of mathematics.

axiomatic formalism (Figure 10.12). Even though logic is here represented as a higher level of symbolism, it should be remembered that Hilbert's reformulation of Euclidean geometry is a verbal and logical extension of the enactive and iconic development of geometry.

5.1 *Reviewing the Data in Terms of Natural and Formal*

An alternative two-part analysis focuses on the development from natural mathematics based on embodiment and symbolism to formal mathematics based on set-theoretic definition and formal proof.

Our concept images are based on our previous experience, including both mental pictures and symbolic processes. Natural thinking builds from natural experiences using thought experiments and symbolic operations. Formal thinking arises from choosing quantified axioms and definitions selected to specify a particular formal situation, constructing a sequence of theorems by formal deduction from the axioms, definitions and previously proven theorems. (Figure 10.13.)

Emotional aspects enter into the mix, as individuals with confidence in their ability to face new ideas may build either naturally from their previous experience or formally in the new axiomatic formal framework. Meanwhile, students who find mathematics problematic and lack confidence may find the subject a source of anxiety and either lose heart or seek to avoid the anti-goal of failure by focusing on the alternate goal of learning procedurally to be able to reproduce proofs in an examination.

There is a huge difference between the embodiment and symbolism of school mathematics and axiomatic formal thinking at university. As a

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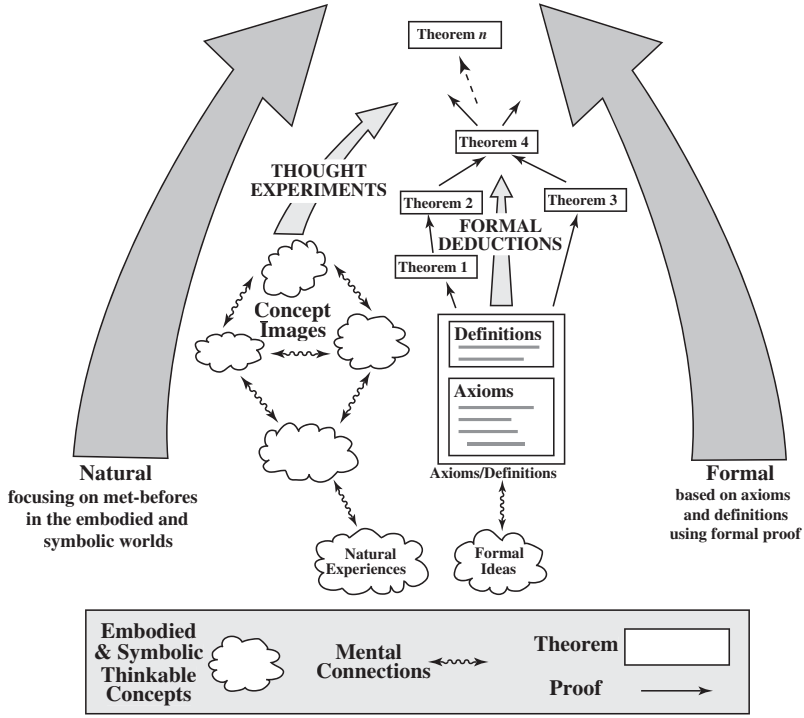

Figure 10.13. Natural and formal routes to build proofs.

university teacher and examiner over many years, I have observed in many situations how students prefer problems that are solvable by well-rehearsed algorithms rather than the subtleties of formal proof.

As an awarder prescribing the grades on a higher mathematics paper taken only by exceptional eighteen-year-olds, the candidates almost unanimously selected long and technical algorithms in the calculus rather than elementary formal proofs in group theory that could be answered in a few lines. In a first-year mathematical analysis course, students preferred to use tests of convergence involving calculations rather than problems requiring proofs. In linear algebra, students are more likely to answer questions solving linear equations. In group theory they preferred to tackle problems involving symbolic manipulations such as operating with permutations. In complex analysis, the first choice usually combines visual and symbolic methods in contour integration. In every case, the students preferred 'safe' questions using routines that they had practiced as an extension of the symbolic world of calculation and manipulation rather than the transition into

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axiomatic proof. Questions requiring even minimal problem solving in formal theory are less likely to be attempted if they involve more than simply reproducing a theorem learnt by rote.

This emphasizes the enormous gulf between the parallel worlds of embodiment and symbolism and the subsequent axiomatic formal world. It is for this reason that I retain the three-part structure of the three worlds of mathematics rather than reduce it to fewer categories. It reflects the major change in mathematical thinking that distinguishes the more fundamental natural worlds of embodiment and symbolism from the distinct axiomatic formal world used for communication of research results by research mathematicians.

6. The Bigger Picture

The data used for the building of the framework has been based mainly on the development of mathematical analysis. However, in Chapter 8 and in the earlier part of this chapter, other aspects of the development of mathematical thinking have been considered that offer a variety of different contexts that combine embodiment, symbolism and formalism in various ways.

Hilbert's formalism takes the formal embodiment of Euclidean geometry to set-theoretic formalism that refers only to the relationship between formal entities consisting of 'points', 'lines' and 'planes' that he observed could just as easily be called 'tables', 'chairs' and 'beermats'. This builds from embodiment to formalism without any reference to the operational symbolism of measurement.

The proof of the area of a spherical triangle in Chapter 8 is an essential blend of embodiment of the sphere and the operational symbolism of the computation of surface area. This is a theoretical argument that does not need to be translated into an axiomatic formal proof.

A variety of possible developments of mathematical proof is outlined in Figure 10.14.

Natural proof builds on concept imagery involving embodiment and symbolism, which may build on embodiment, symbolism, or a blend of the two. A default form of proof occurs through a procedural development that essentially builds on rote-learning proofs for reproduction in examinations. This should not be confused with individuals who commit proofs to memory so that they can reproduce them and reflect upon their content to make sense of them. Formal proof involves making coherent deductions based on set-theoretic definitions.

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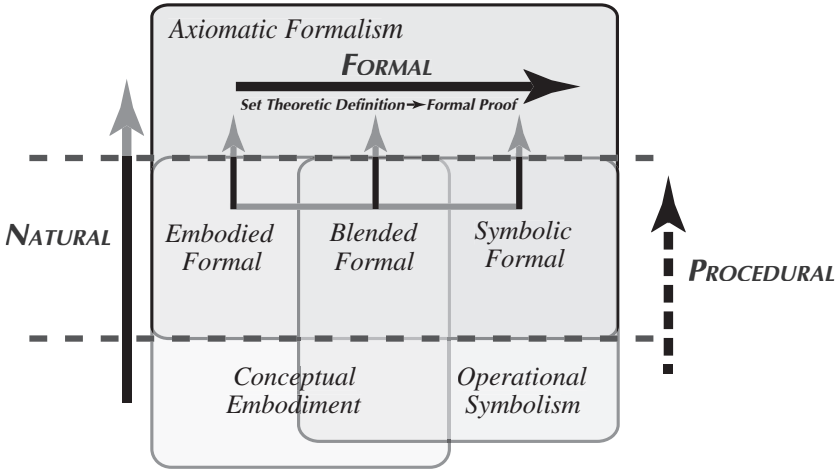

Figure 10.14. The development of proof in embodiment, symbolism and formalism.

The vertical arrows representing the development from natural to formal are shaded black in the lower portion and grey at the top. This represents the observation that proofs often remain in the theoretical world of natural embodiment and symbolism without shifting to the formal world of set-theoretic definition and proof.

Most of us work with natural embodiment and symbolism. Many professional mathematicians are happy to accept such proofs on the understanding that they *could* be expressed in a more formal manner, should that be desired.

Formal proof becomes essential to ensure (as best we can) that the proofs that we present are not dependent on hidden assumptions that may later become problematic. Although natural proof depends on the specific context and may be generalized theoretically, formal proof represents a distinct shift to basing mathematical thinking on quantified reasoning that applies to *any* context that satisfies the set-theoretic axioms and definitions.

7. Reflections

The three-world framework reveals the development from the theoretical mathematics of perception and operation to the formal mathematics of definition and deduction to be a huge challenge for students. Formal knowledge structures can be developed in a spectrum of different ways, from natural approaches using embodied and symbolic experience that give

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meaning to definitions and relationships, to formal approaches extracting meaning from the definition using formal proof.

Students have varying preferences that involve both cognitive and emotional aspects. They may gain confidence or become disaffected through making, or failing to make, sense of a natural or formal approach.

In Chapter 11, we will use the framework from embodiment and symbolism to formalism to explore the cognitive development of the calculus and mathematical analysis. We will then continue the journey building formal knowledge structures in more advanced mathematical thinking.

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